

Test-Set Geographic Composition Dominates Method Ranking in Remote Sensing Super-Resolution Evaluation

Zhengyang Chen, Dawei Liu*, and Zhiheng Liu

Abstract—Remote sensing super-resolution is evaluated on a small number of geographically distinct sites, and methods are ranked by differences of one or two tenths of a decibel. We ask how much of such a ranking is decided by which sites happen to be in the test set. Three multi-temporal Sentinel-2 networks, three seeds each, are evaluated on all 18 sites of the S2DS benchmark separately, and every possible k -site test subset is enumerated. The two strongest networks differ by 0.16 dB on average and are statistically indistinguishable at the site level (Wilcoxon $p = 0.32$, bootstrap 95% CI $[-0.26, +0.59]$ dB). Over the 8568 five-site subsets the better network wins in 67.6% of them, and the benchmark’s own three partitions produce three different orderings of the same three architectures. Site variance exceeds model variance by 31.7 times, and the PSNR range across sites is 11.38 dB, which is 57 times the gap being resolved. A stable ranking requires at least 16 sites. The effect replicates on MuS2, a public benchmark with a different sensor pair and scale factor, where zero-training baselines still span 6.31 dB across seven tiles.

Index Terms—Super-resolution, evaluation protocol, multi-temporal Sentinel-2, benchmark design, geospatial sampling.

I. INTRODUCTION

Super-resolution (SR) for remote sensing is evaluated the way natural-image SR is evaluated [1], [2]. A fixed test split is assembled, every method is run over it, and a single mean PSNR is reported. The architecture that wins by a tenth of a decibel is described as the better one.

This convention carries an assumption that is rarely examined, namely that the test split is large and varied enough for its particular composition not to decide the outcome. In natural-image SR the assumption is defensible, because benchmark splits contain hundreds or thousands of images drawn from the same photographic distribution. Remote sensing data are different. A benchmark for satellite SR is assembled from a small number of geographical sites [3]. Each site is internally homogeneous, and the sites differ from one another in climate, terrain, land cover and atmospheric conditions. The number of independent geographical units is often five or fewer.

Two properties follow. Images from one site are spatially correlated, so samples drawn from a single site are not independent draws from a distribution over scenes. The difficulty of the task itself varies with the site, because texture, contrast and cloud frequency are governed by geography. Both properties push in the same direction, and the number of sites determines how much of that dependence is averaged out.

We test this directly. Rather than treating the benchmark as a single number, we evaluate every model on every site

separately and ask what happens to the rankings as the site composition of the test set changes. The experiment uses nine existing checkpoints.

The contributions are the following. We quantify site-level dispersion in a multi-temporal Sentinel-2 SR benchmark, where the PSNR across 18 sites spans 11.38 dB while the two strongest networks differ by 0.16 dB. We decompose variance between sites and between models, finding a ratio of 31.7. We enumerate every five-site test subset and show that the ranking of the two strongest networks flips in 32.9% of them, and that at least 16 sites are needed for stability. We replicate the effect on MuS2, a benchmark built by others with a different sensor pair and scale factor. Finally, we identify a second way a benchmark can mislead, in that the fraction of valid target pixels falls as low as 1.5% in one site.

II. DATA AND METHODS

A. S2DS Benchmark

S2DS is a multi-temporal SR benchmark built from real Sentinel-2 Level-2A acquisitions [4], [5]. It contains 18 sites partitioned by geography into 10 training, 3 validation and 5 test sites. Each sample provides $T = 12$ low-resolution frames acquired on distinct dates at 40 m, the bands B02, B03, B04 and B08, per-frame per-pixel cloud fraction and per-frame signed time offsets, together with a 10 m target image for the reference date. The scale factor is 4. The split is by site and not by patch, so training and test samples come from disjoint areas. The evaluated splits contain 873, 180 and 423 samples respectively.

B. Models

Three architecture families are used, each with three seeds. Mamba-MISR carries inter-frame interaction in a selective state-space module [6] and gates fusion by per-pixel observation reliability, with temporal offsets entering the gate only. HighRes-Net-cld is the recursive fusion architecture of [7] extended with cloud-mask conditioning. BreizhSR is a lightweight multi-frame network [8] used as a lower-capacity reference. All twelve checkpoints are fixed at 30 000 iterations, with seeds 2026, 2027 and 2028.

C. Evaluation Protocol

PSNR is computed over the valid mask, defined as pixels whose target-date cloud label is clear [9], following the

standard image-quality measures of [10]. Per-site values are unweighted means over the samples of that site. The two aggregation conventions differ, and we state which is used at each point, because on the test split the sample-weighted mean is 27.2285 dB for Mamba-MISR seed 2026 while averaging the five site means gives 27.0993 dB, a difference of 0.1293 dB arising from unequal sample counts. Samples with an empty valid mask are excluded rather than counted as zero, leaving 1314 of 1476 samples for the network comparison.

D. Analysis Design

Every model–seed combination is evaluated separately on each of the 18 sites, giving a 9×18 matrix of site-level PSNR. Variance across site means and variance across model–seed means are computed from that matrix, and the ratio is reported with a bootstrap 95% confidence interval obtained by resampling both sites and model–seed combinations over 2000 replicates.

For a pair of networks the win rate of the first is computed by exact enumeration of every subset of size k drawn from the 18 sites, for k from 3 to 18. With $\binom{18}{5} = 8568$ the enumeration is exact and no sampling approximation is used. The paired site-level difference is tested with a Wilcoxon signed-rank test [11], with the rank-biserial correlation as effect size and a bootstrap confidence interval over the 18 sites. The unit of analysis is the site and not the sample, because samples within one site are not independent. Means and medians are reported together, since the distribution of site-level differences is skewed [12].

E. External Replication on MuS2

MuS2 [3] is a public multi-image SR benchmark built by others from Sentinel-2 as the low-resolution source and higher-resolution multispectral imagery as the reference, with a scale factor of 3. The local copy holds 91 scenes over 7 MGRS tiles.

Two settings are used and reported separately. In Setting A the reference image is downsampled by 3 and reconstructed with nearest, bilinear or bicubic interpolation. Ground truth and prediction then live on the same radiometric scale, so no registration or band-matching uncertainty enters, and the quantity measured is how hard the fabric of a given tile is to reconstruct. In Setting B a Sentinel-2 band is upsampled and compared against the reference band, after a quantile match that maps the prediction onto the reference scale. Registration between the two products is not independently verified, so the Setting B numbers carry that uncertainty and are quoted alongside Setting A rather than on their own.

III. RESULTS

A. Site-Level Dispersion

Fig. 1 shows the valid PSNR of Mamba-MISR averaged over nine model–seed combinations for each of the 18 sites. The values span 11.38 dB, from 18.56 dB at qingdao_inland to 29.94 dB at sjz_north_agri, with

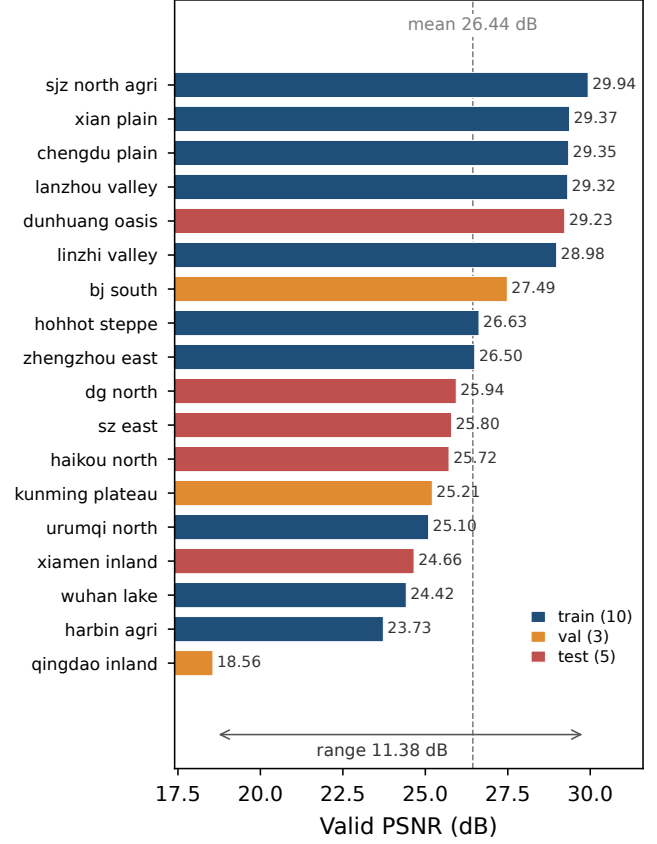


Fig. 1. Valid PSNR at each of the 18 S2DS sites, averaged over nine model–seed combinations and sorted by score. Colour marks the training, validation and test partitions. Site identifiers follow Table I. The 11.38 dB span across sites exceeds the 0.16 dB gap between the two strongest networks by a factor of 71.

a standard deviation of 2.81 dB. The three partition memberships behave as one population. The 10 training sites average 27.33 dB, the 3 validation sites 23.75 dB and the 5 test sites 26.27 dB, and the highest and lowest sites are not separated by partition. *dunhuang_oasis*, a test site, ranks third, while *harbin_agri*, a training site, ranks second from the bottom. Considered per architecture rather than averaged, the spans across the same sites are 12.45, 13.08 and 10.96 dB, so the dispersion is not an artefact of averaging.

A partition effect does exist but is far smaller. Evaluating Mamba-MISR seed 2026 on the training partition gives 27.4640 dB and on the test partition 27.2285 dB, a difference of 0.2355 dB, which is within the range that single-seed training noise produces on this benchmark.

B. Variance Attribution

Variance across site means is 7.886 dB^2 , a standard deviation of 2.81 dB. Variance across model–seed means is 0.249 dB^2 , a standard deviation of 0.50 dB. The ratio is 31.7 with a bootstrap 95% confidence interval of 6.7 to 307. The interval is wide because 18 sites and 9 model–seed combinations give limited resolution at the upper end, but its lower bound already places site-to-site variation an order of magnitude above model-to-model variation. The ratio depends

on which models are compared, and Section III-E shows a public benchmark on which the same computation over four zero-training baselines returns a ratio below one. What is stable across datasets is the site-level dispersion, not the ratio.

C. Ranking Instability

Comparing Mamba-MISR against HighRes-Net-cld, their site-level means over 18 sites are 26.6870 dB and 26.8475 dB. The paired difference has a mean of 0.1605 dB and a median of 0.4474 dB. A Wilcoxon signed-rank test gives $p = 0.3247$ with a rank-biserial correlation of 0.275, and the bootstrap 95% confidence interval of the mean difference is $[-0.2593, +0.5856]$ dB, which contains zero. The two networks are statistically indistinguishable at the site level.

Fig. 2 shows the consequence. Enumerating all 8568 five-site subsets, HighRes-Net-cld scores higher in 67.6% of them, so Mamba-MISR wins 32.9%. A study drawing a different five sites would have reported the opposite ordering in one case out of three. The comparison is informative because it sits at a specific gap size. Two further pairs on the same matrix differ by far more. Mamba-MISR exceeds BreizhSR by 0.90 dB and wins in 99.43% of five-site subsets, and HighRes-Net-cld exceeds BreizhSR by 1.06 dB and wins in 99.03%. A gap near 0.16 dB produces a win rate near one third, while gaps near one decibel produce rates above 99%, which locates the transition region.

The absolute level is equally sensitive. Drawing five sites at random, the mean PSNR of the nine model-seed combinations ranges from 23.40 to 29.94 dB, with a standard deviation of 1.14 dB and a span of 6.54 dB. Raising the subset size to ten narrows the standard deviation to 0.62 dB.

A third view comes from the partition the benchmark defines, which was fixed before this analysis. Ranking the three architectures within each partition gives three different orderings, as Table I shows. On the 10 training sites HighRes-Net-cld leads with 28.20 dB against 27.29 dB. On the 5 test sites the order reverses, with Mamba-MISR at 27.06 dB against 26.11 dB. On the 3 validation sites Mamba-MISR leads at 24.07 dB while HighRes-Net-cld falls to last place at 23.56 dB, below BreizhSR at 23.64 dB. The validation and test partitions are equally unseen by all three models, so the reversal between them is attributable to which sites they contain and to nothing else.

D. Scene-Level View

The same comparison at scene level, over the 1314 samples that retain a non-empty valid mask for both networks, gives Mamba-MISR the better score in 36.6% of scenes and a 0.195 dB loss on average (Fig. 3). The direction and magnitude agree with the site-level result. Stratifying scenes by the PSNR of Mamba-MISR does not explain the pattern, since the win rate by quartile, from hardest to easiest, is 0.343, 0.366, 0.424 and 0.331 and the correlation with scene difficulty is 0.17.

E. How Many Sites Are Enough, and External Replication

Fig. 2 also traces the win rate as the subset size grows. The values for Mamba-MISR against HighRes-Net-cld are 0.390 at

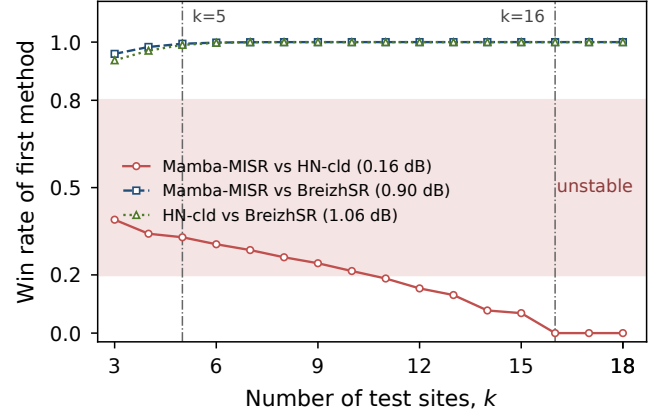


Fig. 2. Win rate of the first method as the number of test sites k grows, by exact enumeration of all $\binom{18}{k}$ subsets. The shaded band marks the region in which the ranking is unstable. A gap of 0.16 dB leaves the ordering undecided until $k = 16$, while gaps near one decibel are stable from $k = 3$.

TABLE I
VALID PSNR (dB) AT EACH S2DS SITE, MEAN OVER THREE SEEDS

Site	Split	MISR	HN-cld	BreizhSR
sjz north agri	Tr	29.83	31.05	28.93
xian plain	Tr	29.27	30.48	28.37
chengdu plain	Tr	31.09	31.30	25.64
lanzhou valley	Tr	29.16	29.85	28.94
dunhuang oasis	Te	29.78	28.12	29.77
linzhi valley	Tr	28.91	29.62	28.41
bj south	Va	27.84	27.26	27.36
hohhot steppe	Tr	26.15	27.18	26.55
zhengzhou east	Tr	26.13	27.21	26.15
dg north	Te	27.01	26.20	24.61
sz east	Te	26.33	25.82	25.24
haikou north	Te	26.48	25.72	24.95
kunming plateau	Va	25.72	25.18	24.74
urumqi north	Tr	24.68	25.77	24.84
xiamen inland	Te	25.70	24.69	23.59
wuhan lake	Tr	24.15	25.36	23.76
harbin agri	Tr	23.51	24.19	23.49
qingdao inland	Va	18.64	18.23	18.82
Range		12.45	13.08	10.96
Train mean (10)		27.29	28.20	26.51
Val mean (3)		24.07	23.56	23.64
Test mean (5)		27.06	26.11	25.63

$k = 3$, 0.329 at $k = 5$, 0.261 at $k = 8$, 0.213 at $k = 10$, 0.154 at $k = 12$, 0.069 at $k = 15$ and 0.000 at $k = 16$. The win rate crosses the 5% threshold only at $k = 16$. For a gap of this size twelve sites would still leave a 15% chance of reporting the wrong winner, and the five-site test sets in routine use sit far outside the stable region.

The replication uses a benchmark built by others, from a different sensor pair, at a different scale factor, and with no model training at all. Under Setting A the PSNR range across the seven MGRS tiles is 5.54 dB for nearest-neighbour, 6.92 dB for bicubic and 6.97 dB for bilinear, as Fig. 4(a) shows, and the pattern holds at both wavelengths tested. Averaged over the six Setting A baselines the range is 6.31 dB. Under Setting B the range is 14.13 and 8.77 dB for the two bands, averaging 11.45 dB.

Fig. 4(b) places these ranges beside the S2DS value of 11.38 dB and beside the 0.05 to 0.20 dB band occupied by typical method gaps. Setting A, which carries no registration uncertainty, sits at 32 times the upper end of that band, and Setting B and S2DS sit at 57 times.

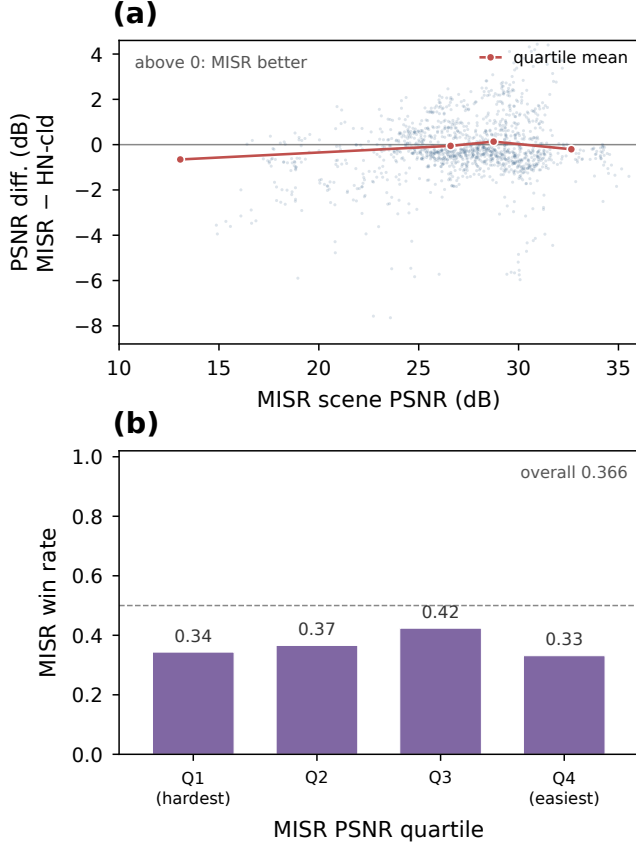


Fig. 3. (a) Per-scene PSNR difference against the PSNR of Mamba-MISR, with quartile means overlaid. (b) Mamba-MISR win rate by difficulty quartile. The advantage is not concentrated in easy or hard scenes.

Site variance on MuS2 has a standard deviation of 2.87 dB in band b3 and 2.70 dB in b6, closely matching the 2.81 dB of S2DS. The two bands are analysed separately because they differ by 5 to 8 dB in absolute level.

F. Valid-Pixel Fraction

A second and independent way a benchmark can mislead emerged during the site-level analysis. Fig. 5 shows that the fraction of valid, cloud-free pixels in the target image varies by a factor of seven across the 18 sites, from 0.147 to 0.994. The scene-level distribution is more extreme. At qingdao_inland the median valid fraction is 0.015, so more than half of its scenes have almost no usable pixels, and its PSNR is the lowest of the 18 sites at 18.56 dB. The relationship between valid fraction and PSNR is positive but partial, with a Pearson correlation across sites of 0.54, and qingdao_inland sits six to eight decibels below the trend. The valid fraction accounts for part of the site's weakness and not all of it, and the remaining cause is not identified here. What matters for reporting is that a mean over 1.5% of a scene carries a variance that no number of scenes removes when the same acquisition conditions recur.

IV. DISCUSSION

The two strongest networks in this study differ by 0.16 dB. Their site-level paired difference is not distinguishable from

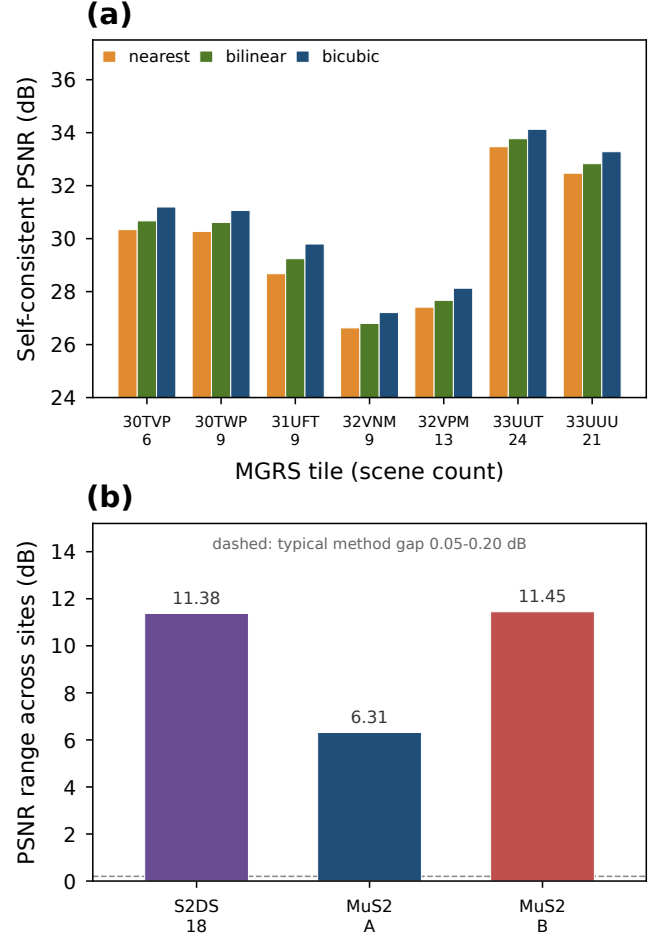


Fig. 4. (a) Self-consistent PSNR of three zero-training baselines at each of the seven MuS2 tiles. (b) Range across sites for S2DS, MuS2 Setting A and MuS2 Setting B, against the 0.05–0.20 dB band occupied by typical method gaps.

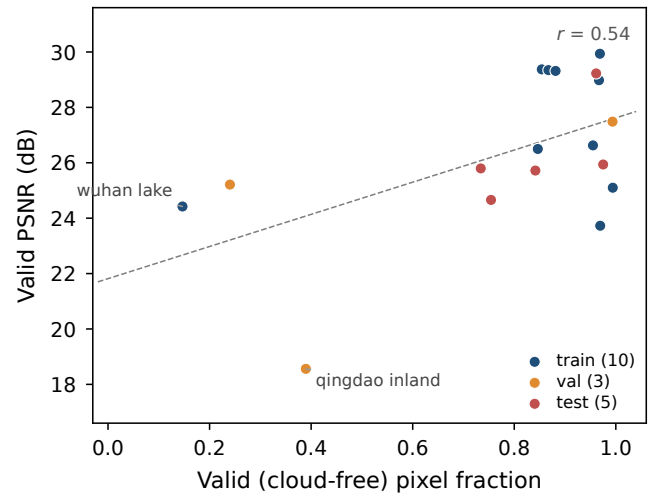


Fig. 5. Valid (cloud-free) pixel fraction against valid PSNR for the 18 sites. The relationship is positive but partial, and the lowest-scoring site sits well below the trend.

zero, and over five-site test subsets the ordering reverses in one case out of three. The same networks vary by 11.38 dB across

sites, and the mean score moves by 6.54 dB depending on which five sites are drawn. The last two figures are one to two orders of magnitude larger than the quantity being resolved. The fragile quantity is the narrow gap, which is the regime that method papers occupy when they report an improvement.

A benchmark score has always been understood to depend on its test data. The quantities reported here give that dependence a magnitude, specific to a benchmark family and to a gap size. How much does the score move when the test sites change. How many sites are needed before a ranking of a given gap size settles. At what gap size does the ranking stop depending on site composition. The answers here are 6.54 dB, 16 sites, and somewhere between 0.16 and 0.90 dB. The MuS2 replication matters for the same reason, since the effect appears on a benchmark built by other authors with different sensors and a different scale factor, and the magnitude differs in the direction that the two benchmarks' geographical extents predict.

Four reporting practices follow. Report per-site scores, since a single aggregate conceals a distribution whose spread exceeds the effect being claimed, and the per-site values cost a few lines of a table. Run leave-one-site-out, which shows whether the headline number survives the removal of any single site. Report rank stability rather than a single ordering, since for two methods within a decibel of each other a win rate over plausible test sets carries more information than a rank. Treat the number of sites as a design parameter, because a test set of three to five geographically distinct sites cannot resolve differences below roughly one decibel on this benchmark. None of the four requires extra training.

V. CONCLUSION

A benchmark score in remote sensing SR is decided to a large extent by which sites the benchmark happens to contain. On a multi-temporal Sentinel-2 benchmark, methods differing by 0.16 dB exchange places in one third of five-site test sets, site-to-site variance exceeds model-to-model variance by 31.7 times, and the benchmark's own three partitions yield three different orderings of the same models. At least 16 sites are needed before a gap of this size becomes resolvable. The effect reproduces on an independently built public benchmark where no model is trained at all. Reporting per-site scores, running leave-one-site-out, and stating the rank stability of narrow gaps would make these limits visible at no experimental cost.

REFERENCES

- [1] A. B. Molini, D. Valsesia, G. Fracastoro, and E. Magli, "DeepSUM: Deep neural network for super-resolution of unregistered multitemporal images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3644–3656, May 2020.
- [2] S. Khanna, P. Liu, L. Zhou, C. Meng, R. Rombach, M. Burke, D. B. Lobell, and S. Ermon, "DiffusionSat: A generative foundation model for satellite imagery," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2024.
- [3] P. Kowalczyk *et al.*, "MuS2: A real-world benchmark for Sentinel-2 multi-image super-resolution," *Sci. Data*, vol. 10, 2023, Art. 712.
- [4] M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, *et al.*, "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," *Remote Sens. Environ.*, vol. 120, pp. 25–36, May 2012.
- [5] ESA, *Sentinel-2 Products Specification Document*, issue 14.9, 2023.

- [6] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," arXiv:2312.00752, Dec. 2023.
- [7] M. Deudon, A. Kalaitzis, S. Gowda, *et al.*, "HighRes-net: Recursive fusion for multi-frame super-resolution of satellite imagery," in *Proc. IEEE/CVF CVPR Workshops*, 2020, pp. 298–307.
- [8] A. Okabayashi, N. Audebert, S. Donike, and C. Pelletier, "Cross-sensor super-resolution of irregularly sampled Sentinel-2 time series," in *Proc. IEEE/CVF CVPR Workshops (EarthVision)*, 2024, pp. 502–511.
- [9] S. Skakun, J. Wevers, C. Brockmann, G. Doxani, *et al.*, "Cloud Mask Intercomparison eXercise (CMIX): An evaluation of cloud masking algorithms for Landsat 8 and Sentinel-2," *Remote Sens. Environ.*, vol. 274, 2022, Art. 112990.
- [10] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
- [11] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, Dec. 1945.
- [12] D. Lakens, "Equivalence tests: A practical primer for *t* tests, correlations, and meta-analyses," *Soc. Psychol. Personal. Sci.*, vol. 8, no. 4, pp. 355–362, 2017.

Manuscript received XXXX; revised XXXX. The authors are with the Engineering University of PAP, Xi'an 710000, China.

*Corresponding author: Dawei Liu (e-mail: wjmicheal@163.com).

Code and the per-site evaluation tables are available at <https://github.com/czy326/s2ds-test-composition>.